

PLANT ALKALOID EFFECT GRID

60 substances × 4 effect classifications · compiled 2026-09-19 · companion to the evidence report (248 sources)

Marks: ● Documented (human studies) ◐ Claimed (traditional/user reports) ✕ Disputed (human evidence negative) · none — ◦ = boundary entry (alkaloid-like or alkaloid unconfirmed). **Tiers:** 1 established safe · 2 borderline with caveats · 3 excluded. Educational reference only — not medical advice; legal status changes rapidly, verify locally.

Substance	Source Plant	Stim	Mood	Calm	Enh	Depend.	Legal (US/EU)	Evidence Note
Tier 1 — Established Safe (8)								
Caffeine	Coffea arabica, Camellia sinensis, Paullinia cupana, Ilex paraguariensis	●	●	·	·	low	Legal worldwide; GRAS (US), EFSA ≤400 mg/day	Gold standard: RCTs show vigor/alertness/mood at 50–300 mg; self-limiting inverted-U (irritability past ~500 mg).
Theobromine	Theobroma cacao	◐	●	·	·	low	Legal; FEMA GRAS flavoring	Subtle: positive affect only at ~250 mg isolated; dysphoric at 500–1000 mg (Baggott 2013, n=84). Chocolate effect ≠ isolated alkaloid.
Capsaicin ◦	Capsicum annuum et al.	·	●	·	·	none	Legal food; GRAS	Chili high: modest but real (SCAN 2025 n=45); endogenous-opioid mediated; extreme-dose harms only in challenge formats. Technically a capsaicinoid amide.
Protopine / allocryptopine	Eschscholzia californica (California poppy)	·	·	●	·	none	Legal supplement US/EU; EMA traditional use	Weak-documented: one positive RCT (n=264, combination product); very low toxicity.
Harman / norharman (β-carbolines)	Passiflora incarnata (passionflower)	·	◐	◐	◐	none	Legal; food-grade herb	Plant-level anxiolysis documented (RCT ≈ oxazepam); alkaloid-vs-flavonoid attribution disputed — alkaloid-level evidence is claimed.
Sanjoينية A (franguloline)	Ziziphus jujuba var. spinosa (suan zao ren)	·	·	◐	·	none	Legal; food-grade	GABAergic anxiolytic/hypnotic in animals; human RCTs only for multi-herb formulas. Food-grade safety.
Trigonelline	Coffea spp., Trigonella foenum-graecum	·	·	·	·	none	Legal; dietary	Safe (2023 formal risk assessment) NAD+ precursor — but no psychoactive effect in humans. Included as the control case: safe ≠ felt.
Piperine	Piper nigrum (black pepper)	·	✕	·	·	none	Legal food; EU 2024 priority-list review	Safe at culinary/supplemental doses, but human mood evidence is null (BJN 2014 RCT) — plasma levels 100–1000× below in vitro potencies. Interaction liability is the real hazard.
Tier 2 — Borderline, Caveats (24)								
Paraxanthine	Coffea/Camellia (caffeine metabolite; supplemental)	●	◐	·	·	low	Self-affirmed GRAS (US, 2021)	3 small industry-funded RCTs (n=12–13, ≤7 days) show attention benefits; no independent replication; raises striatal dopamine in rats unlike caffeine.
Theacrine	Camellia assamica var. kucha	●	◐	·	·	low	Legal supplement (US); EU novel-food status unsettled	Small (partly industry-linked) human trials: energy without tolerance signal at ≤300 mg; long-term data thin.
Methylxerine (Dynamine)	Camellia kucha, Coffea spp.	●	◐	·	·	low	Self-affirmed GRAS (US)	Small RCTs: no HR/BP effect at 100 mg; mood claims vendor-driven; long-term data absent.
Mesembrine / mesembrenone (kanna)	Sceletium tortuosum	·	●	●	◐	low	Unscheduled US; EU novel-food authorization proposed Nov 2025 (25 mg/day)	Best-documented traditional mood alkaloid — but 2023 meta-analysis (4 RCTs, n=117) found no anxiety benefit vs placebo. All positive trials tied to one proprietary extract.
Mitragynine (kratom leaf)	Mitragyna speciosa	●	◐	◐	·	moderate	US: unscheduled but FDA-deemed unlawful in supplements; ~24 states regulate, several ban; banned in parts of EU	Low-dose stimulant, high-dose opioid-like. Survey benefits (pain 91%, n=2798) but zero efficacy RCTs; dependence/withdrawal documented. Leaf vs concentrate distinction is decisive.
Galantamine	Galanthus spp. (snowdrop)	·	·	·	●	none	Prescription drug (US/EU); supplement status contested	Lucid dreaming RCT (LaBerge 2018, n=121): 42% vs 14% placebo at 8 mg; dose-dependent. Genuine enhanced-experience evidence, drug-status caveat.
Huperzine A	Huperzia serrata (club moss)	·	·	·	◐	none	Legal supplement (US)	Acceptable short-term safety (≤200 µg/day); cognitive agent — dream/enhancement claims mechanistic extrapolation, no human subjective-effect trials.
Erythravine (mulungu)	Erythrina mulungu	·	◐	●	·	none	Legal supplement US/EU	Anxiolysis mixed: one positive small human crossover, one null. Diazepam-like in rodents. Embryo-fetal toxicity signal in rats — avoid in pregnancy.
Rhynchophylline / isorhynchophylline	Uncaria rhynchophylla (gou teng)	·	◐	◐	·	none	Legal; TCM herb	Rodent antidepressant/anxiolytic via MAO/5-HT1A; no human trials for subjective effects.
Neferine / liensinine	Nelumbo nucifera embryo (lotus plumule)	·	◐	◐	·	none	Legal food/herb	Animal sedation/antidepressant-like effects ≈ imipramine in forced swim; zero human trials. Food-safe history.
Sinomenine	Sinomenium acutum	·	◐	◐	·	low	Prescription drug in China; supplement elsewhere	Multiple mouse depression models (BDNF); no human mood trials; documented histamine-release reactions.

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Tetrahydropalmatine (l-THP, rotundine)	<i>Corydalis yanhusuo</i>	•	✗	●	•	low	Drug in Chinese pharmacopoeia since 1977; unscheduled US	Documented for tension insomnia (D1/D2 antagonist + GABA-A) — but anti-reward pharmacology means mood-lift marketing is misplaced. Rare hepatotoxicity case reports.
Leonurine	<i>Leonurus cardiaca</i> (motherwort)	•	•	●	•	none	Legal herb	Uncontrolled human pilot on herb positive; alkaloid-level data animal-only. Otherwise low toxicity.
Akuammine group	<i>Picralima nitida</i> (akuamma)	•	●	●	•	low	Unscheduled US; legal	Traditional analgesic/calming use; vendor-driven euphoria claims; no human trials. μ/k-opioid activity warrants caution.
Mitraphylline / mitrajavine	<i>Mitragyna hirsuta</i> , <i>M. javanica</i>	●	●	•	•	low	Legal where kratom banned (gray zone)	Vendor-claimed 'kratom-lite' mood/stimulation; presumed milder; almost no human data.
Cytisine	<i>Cytisus laburnum</i> , <i>Sophora</i> spp.	●	●	•	•	low	OTC cessation drug E. Europe/UK/Canada; not FDA-approved	Strong rodent antidepressant data; human use is smoking cessation only — mood/energy claims are extrapolation.
Synephrine	<i>Citrus aurantium</i> (bitter orange)	✗	•	•	•	none	Legal supplement; ANSES cap <20 mg/day	~30 human studies: no stimulant effect at ≤100 mg/day — the marketed 'energy' effect is disputed; chronic use raises BP ~+6/+4 mmHg (2022 meta).
Yohimbine	<i>Pausinystalia johimbe</i>	●	✗	•	•	low	Rx-only UK/Canada/Australia; chaotic US supplements (0–12.1 mg/serving)	Alpha-2 antagonist; effective for ED but a standard laboratory panicogen (~60% panic rate in PTSD challenge). Documented fatal overdoses.
Nuciferine (lotus leaf)	<i>Nelumbo nucifera</i> leaf	•	•	●	●	none	Legal herb (Chinese pharmacopoeia); blue lotus DoD-prohibited	Traditional 'calming the mind' use + preclinical GABA-A anxiolysis — but aripiprazole-like profile and adverse US military case series (concentrated/vaped products) keep heavy caveats.
Calea zacatechichi actives ◦	<i>Calea zacatechichi</i> (Mexican dream herb)	•	•	●	●	none	Uncontrolled US/EU	One small double-blind human study (Mayagoitia 1986) supports dream enhancement. Actives are sesquiterpene lactones; the reported alkaloid was never characterized — boundary entry.
Silene capensis β-carbolines	<i>Silene capensis/undulata</i> (African dream root)	•	•	•	●	none	Uncontrolled	Ethnographic oneirogen; 2024 LC-MS/MS detected β-carbolines + 5-HT2A docking — in silico only, no human trials.
Entada rheedii actives ◦	<i>Entada rheedii</i> (African dream bean)	•	•	•	●	none	Uncontrolled	Traditional dream-enhancing use; tryptophan-derivative chemistry, alkaloid status unclear — boundary entry.
Blue lotus aporphine fraction	<i>Nymphaea caerulea</i>	•	●	●	●	low	Unscheduled US except Louisiana; DoD-prohibited	Military case series shows real CNS activity (agitation, tachycardia as adverse events); traditional calming use; no controlled human data.
Berberine	<i>Berberis</i> spp., <i>Coptis chinensis</i>	•	●	•	•	none	Legal supplement	Mood signal is one IBS-D RCT's depression/anxiety subscales — likely secondary to GI relief. Metabolic agent, not a feel-good agent.
Tier 3 — Excluded (Educational) (28)								
Nicotine	<i>Nicotiana tabacum</i>	●	●	●	•	high	Legal adult product worldwide; heavily regulated	Fails on addiction, not acute toxicity: highest dependence capture rate (~32%); delivery format kills 8M/yr globally.
Ephedrine	<i>Ephedra sinica</i> (ma huang)	●	●	•	•	moderate	Supplements banned US (2004); drug behind-counter	~155 deaths led to FDA ban; <1% of supplement sales caused 64% of serious supplement adverse events.
Cocaine	<i>Erythroxylum coca</i>	●	●	•	•	high	Schedule II US; UN 1961 Schedule I	Intense euphoria inseparable from cardiotoxicity; 29,449 US overdose deaths 2023.
Morphine / opium alkaloids	<i>Papaver somniferum</i>	•	●	●	•	high	Schedule II US (essential medicine); heroin Schedule I	Euphoria and lethal respiratory depression are the same mechanism; 81,083 opioid-involved US deaths 2023.
Arecoline	<i>Areca catechu</i> (betel nut)	●	●	•	•	high	Unregulated most countries (Turkey: narcotic)	IARC Group 1 carcinogen; pooled oral-cancer RR 7.9; true dependence syndrome. The template case of positive effect + catastrophic safety failure.
Cathinone / cathine (khat)	<i>Catha edulis</i>	●	●	•	•	moderate	Schedule I US (cathinone); banned UK/most EU	Documented euphoria/energy in human studies; dependence and cardiac/psychiatric harms; cathine Schedule IV.
7-Hydroxymitragynine (concentrated)	<i>Mitragyna speciosa</i> (semi-synthetic concentrates)	•	●	●	•	high	FDA recommended Schedule I Jul 2025; DEA orders pending 2026	13× morphine functional potency; naloxone-reversible respiratory depression in rats; poison-center coma/ICU cases since Feb 2025. The concentration hazard case.
Strychnine	<i>Strychnos nux-vomica</i>	●	•	•	•	none	Restricted poison	Documented CNS-stimulant pharmacology and historical tonic use — but a lethal convulsant with a razor-thin margin. Included as the extreme counter-example.
Higenamine	<i>Nandina</i> , <i>Nelumbo</i> , <i>Aconitum</i>	●	●	•	•	none	FDA: not a lawful supplement ingredient; WADA-banned	Real β2 pharmacology; 'feel-good stimulant' claims vendor-only; cardiovascular harms documented.
Lobeline	<i>Lobelia inflata</i>	●	✗	•	•	low	Legal herb; FDA banned cessation products 1993	Nicotinic pharmacology documented, but human cessation/mood trials negative and margin is narrow — disputed claims.

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Ibogaine	<i>Tabernanthe iboga</i>	•	●	•	●	none	Schedule I US	≥33 publicly reported deaths incl. at therapeutic doses without cardiac history; NIDA abandoned research over cardiotoxicity.
Psilocybin / psilocin	<i>Psilocybe</i> spp.	•	●	•	●	none	Schedule I US; carve-outs OR/CO, Australia 2023	Physiologically very safe but psychological risk real: 39% rate worst trip top-5 most challenging; 7.6% sought treatment for lasting symptoms (Carbonaro 2016). Most contestable exclusion.
Mescaline	<i>Lophophora williamsii</i> , <i>Trichocereus</i> spp.	•	●	•	●	none	Schedule I US (AIRFA tribal exemption)	No documented addiction; intoxications generally mild — excluded on psychosis/panic risk, adulteration, and Schedule I status.
DMT / 5-MeO-DMT	<i>Psychotria</i> / <i>Anadenanthera</i> / <i>Virola</i> spp.	•	●	•	●	none	Schedule I US (UDV/Gospel exemptions)	5-MeO-DMT observational mood improvement documented; ayahuasca MAOI combinations add interaction hazards.
Harmine / harmaline	<i>Banisteriopsis caapi</i> , <i>Peganum harmala</i>	•	●	•	●	none	Unscheduled plants US; DMT admixtures Schedule I	Ayahuasca RCTs show real antidepressant effects — defeated by serotonin-syndrome risk, narrow dosing margin, abortifacient properties.
Ergine / LSA	<i>Ipomoea tricolor</i> , <i>Turbina corymbosa</i> , <i>Argyreia nervosa</i>	•	✗	•	●	none	Schedule III US; seeds legal to possess	Psychoactive documented but euphoria inconsistent — dysphoria/nausea common. Fails mood class on balance.
Bufotenine	<i>Anadenanthera</i> spp. (<i>yopo</i> / <i>vilca</i>)	•	✗	•	✗	none	Schedule I US	Human IV studies unpleasant; intranasal null (Isbell) — snuff effects likely attributable to DMT/5-MeO-DMT. Disputed on both classes.
Cryogenine / vertine (sinicuichi)	<i>Heimia salicifolia</i>	•	✗	•	✗	none	Uncontrolled	Isolated-alkaloid human experiments negative (Malone & Rother 1994) — famed auditory effects failed replication; chronic-use memory impairment reported.
Reserpine	<i>Rauvolfia serpentina</i>	•	✗	●	•	none	Prescription; largely withdrawn	The negative control: documented sedation but causes depression/suicidality via monoamine depletion — calm ≠ positive.
Gelsemine	<i>Gelsemium sempervirens</i>	•	•	●	•	none	Homeopathic dilutions legal; herb toxic	Animal anxiolysis claims but lethal narrow margin (LD50 0.2 mg/kg class) — excluded on toxicity.
Chelidonine	<i>Chelidonium majus</i> (greater celandine)	•	•	●	•	none	EMA negative assessment	Claimed sedative; documented hepatotoxicity — excluded.
Voacangine / voacamine	<i>Voacanga africana</i>	●	●	•	●	none	Unscheduled US; precursor-monitored	Traditional stimulant/visionary use; iboga-class cardiotoxicity concern; poorly characterized.
Vinpocetine	<i>Vinca minor</i> (semi-synthetic)	•	✗	•	•	none	FDA 2019 pregnancy warning; non-dietary-ingredient status	No subjective drug effects in healthy volunteers (Subhan & Hindmarch 1985) — fails both prongs.
Evodiamine / rutaecarpine	<i>Tetradium ruticarpum</i> (<i>Evodia</i>)	✗	•	•	•	none	Gray-zone supplement	Stimulant/thermogenic marketing rests on rodent data; documented hepato-/cardiotoxicity signals.
Hordenine	<i>Hordeum vulgare</i> (barley), beer	✗	•	•	•	none	FDA: supplements adulterated; DoD/NCAA prohibited	Mechanism-hype anatomy: in vitro D2 Ki 13 μM vs human beer plasma 12–17 nM — 1000× too low. 1894 self-experiment: no effect.
Tyramine	<i>Fermented/aged foods</i>	✗	•	•	•	none	Legal dietary component	Pure pressor amine; no positive subjective effects; dangerous with MAOIs.
Conessine	<i>Holarrhena pubescens</i> (<i>kurchi</i>)	•	•	•	✗	none	Not marketed	Cardiotoxic; causes vertigo/tremors/agitation — no positive subjective effects. Screened and rejected.
Theophylline	<i>Camellia sinensis</i> (trace)	●	•	•	•	low	Prescription drug	Tea trace exposure is inert; pharmaceutical dosing is toxic — fails the casual-consumption criterion.

Classifications: Stimulation/Energy · Mood Elevation · Calm/Anxiety Relief · Enhanced Experience. Effect claims marked ● are traditional or user-reported and have not been confirmed in human trials. Update source: data.js in the interactive web version.